

The Seed's Big Journey

I PUC · Ch 5 · Morphology of Flowering Plants

Starring Beeja the seed 🌱 & Buzz the bee 🐝

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Waiting Under the Soil

A tiny bean seed named Beeja lay under the soil, waiting for rain. "What am I made of?" she wondered. Outside was her **seed coat** with two layers — the outer **testa** and the inner **tegmen**. A scar called the **hilum** showed where she was once attached to the fruit, and a tiny pore, the **micropyle**, sat just above it.

💡 **REMEMBER**

A dicot seed has a seed coat, two fleshy cotyledons and an embryonal axis with a radicle and a plumule.



The First Drink

One morning rain soaked the soil. Water rushed in through the micropyle and Beeja swelled up — **germination** had begun! Inside, two thick **cotyledons** fed the **embryonal axis**. Her neighbour, a maize grain, explained, "I am a monocot. My food is in the **endosperm**, covered by a protein layer called the **aleurone layer**. My single cotyledon is the **scutellum**."

💡 **REMEMBER**

Maize: plumule is covered by the coleoptile and radicle by the coleorhiza. Most monocot seeds are endospermic.



Roots Go Down

First the **radicle** pushed downward and became the main root. Its branches formed a **tap root system**, typical of dicots like mustard. The maize grain grew a bunch of thin roots instead — a **fibrous root system**. "Some roots even grow from stems," said an earthworm, pointing at a banyan tree whose **prop roots** hung down like pillars. These are **adventitious roots**.

💡 **REMEMBER**

Root regions from tip: root cap, region of meristematic activity, region of elongation and region of maturation (root hairs).



The Shoot Rises

Now the **plumule** rose toward the sunlight and became the stem. Beeja's stem had **nodes**, where leaves are born, and **internodes** between them. Water climbed up through thin pipes called **xylem**, while food travelled through **phloem**. A pumpkin vine waved its **tendrils** — "My stem is modified for climbing!" — and a citrus bush showed its sharp **thorns**.

💡 **REMEMBER**

Stem modifications: tendrils (cucumber, grapevine), thorns (Citrus, Bougainvillea), underground stems for storage (potato, ginger).



Leaves Unfold

Green leaves opened at each node. Each had a **leaf base**, a stalk called the **petiole** and a flat **lamina**. Beeja's veins formed a net — **reticulate venation** — while the maize leaf had straight lines: **parallel venation**. The leaves were arranged in a pattern called **phyllotaxy**: alternate in Beeja, opposite in guava, whorled in Alstonia.

💡 **REMEMBER**

A simple leaf has an undivided lamina; a compound leaf has leaflets — pinnately compound (neem) or palmately compound (silk cotton).



Time to Flower

Weeks later Beeja grew buds. They were arranged on an axis — the **inflorescence**. In a **racemose** inflorescence the main axis keeps growing; in a **cymose** one it ends in a flower. Each flower had four whorls: **calyx**, **corolla**, **androecium** and **gynoecium**. Her ovary sat above the other parts, so she was **hypogynous**.

💡 **REMEMBER**

Perigynous: ovary half-inferior (plum, rose). Epigynous: ovary inferior (guava, cucumber, ray florets of sunflower).



Calling the Pollinators

Bright petals and sweet scent called Buzz the bee. "Your petals overlap in a special way," buzzed Buzz — the **vexillary** aestivation of the pea family. He brushed against the **stamens**, each made of a **filament** and an **anther**, and carried pollen to the **stigma**. Inside the ovary, ovules sat on a ridge called the **placenta** — **marginal placentation**.

💡 **REMEMBER**

Types of placentation: marginal (pea), axile (china rose, tomato, lemon), parietal (mustard, Argemone), free central (Dianthus) and basal (sunflower).



Seeds for Tomorrow

After fertilisation the ovary became a **fruit** — a pod — and the ovules became new seeds. "A fruit formed without fertilisation is a **parthenocarpic fruit**," buzzed Buzz. Beeja belonged to the family **Fabaceae**: zygomorphic flowers, ten diadelphous stamens and a pod. As the pods dried and split, new Beejas fell to the soil, ready for their own big journey.

💡 **REMEMBER**

Floral formula of Fabaceae: % ♂ K(5) C1+2+(2) A(9)+1 G1. Members: pea, soybean, groundnut, gram.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

seed coat p.1 · testa p.1 · tegmen p.1 · hilum p.1 · micropyle p.1 · germination p.2 · cotyledons p.2 · embryonal axis p.2 · endosperm p.2 · aleurone layer p.2 · scutellum p.2 · radicle p.3 · tap root system p.3 · fibrous root system p.3 · prop roots p.3 · adventitious roots p.3 · plumule p.4 · nodes p.4 · internodes p.4 · xylem p.4 · phloem p.4 · tendrils p.4 · thorns p.4 · leaf base p.5 · petiole p.5 · lamina p.5 · reticulate venation p.5 · parallel venation p.5 · phyllotaxy p.5 · inflorescence p.6 · racemose p.6 · cymose p.6 · calyx p.6 · corolla p.6 · androecium p.6 · gynoecium p.6 · hypogynous p.6 · vexillary p.7 · stamens p.7 · filament p.7 · anther p.7 · stigma p.7 · placenta p.7 · marginal placentation p.7 · fruit p.8 · parthenocarpic fruit p.8 · Fabaceae p.8



QUICK QUIZ

Do you remember?

1. What is germination?

- Formation of flowers
- Growth of the embryo into a seedling
- Fusion of gametes
- Falling of leaves

2. Which tissue transports water in plants?

- Phloem
- Xylem
- Cortex
- Epidermis

3. What is the main function of a flower?

- Photosynthesis
- Storage of food
- Sexual reproduction
- Absorbing water

4. A tap root develops from the:

- Plumule
- Radicle
- Cotyledon
- Stem node

5. After fertilisation the ovary develops into the:

- Seed
- Fruit
- Embryo
- Endosperm

Answers: 1-Growth of the embryo into a seedling · 2-Xylem · 3-Sexual reproduction · 4-Radicle · 5-Fruit